[S08] LoRA and QLoRA: Parameter-Efficient Fine-Tuning of Large Models

TOPIC AREA Modern AI Models

I. Learning Objectives and Guiding Question

Explain low-rank weight updates, quantify trainable-parameter savings, and distinguish LoRA and QLoRA from full fine-tuning.

Guiding question: How can a model with millions or billions of parameters be adapted without updating all its weights?

II. Required Tasks

1. Distinguish pretraining, full fine-tuning, linear probing, adapters, and parameter-efficient fine-tuning (PEFT).
2. Explain low-rank factorization, rank r, and scaling factor α. Identify which parameters are frozen and which are trained.
3. Calculate trainable-parameter counts for a 1024 × 1024 weight matrix using full fine-tuning and LoRA ranks 4, 8, and 16. Estimate the reduction ratios, and distinguish parameter counts from actual GPU memory consumption.
4. Illustrate LoRA training and inference pipelines, including adapter merging; explain intuitively how QLoRA differs from LoRA (a frozen quantized base model plus low-rank adapters).
5. If resources permit, fine-tune a very small model or synthetic layer and compare two ranks. Otherwise, use a toy linear-layer experiment and compare parameter counts with a simple quality measure.
6. Analyze trade-offs among model quality, memory use, training time, inference efficiency, and experimental risks. Describe when LoRA may not be the best choice.

III. Equations / Representations to Explain

$$W' = W + \Delta W,\ \Delta W = (\alpha/r)BA$$

where $W \in \mathbb{R}\hat{}\{d \times k\}, A \in \mathbb{R}\hat{}\{r \times k\}, B \in \mathbb{R}\hat{}\{d \times r\}, r \ll min(d,k)$

Student need to define every symbol, specify appropriate units/value ranges and modeling assumptions, and provide a complete step-by-step worked example.

IV. Minimum Expected Outputs

- Diagram of W + BA.
- A table of trainable parameters across ranks
- Either a toy-experiment result or an annotated workflow.

V. Required and Recommended Research Papers

Required: Hu, E. J. et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.

https://mlanthology.org/iclr/2022/hu2022iclr-lora/

Reading focus: Focus on W + BA, low-rank updates, and trainable-parameter counts.

Recommended: Dettmers, T. et al. (2023). QLoRA: Efficient Finetuning of Quantized LLMs. NeurIPS 2023.

https://mlanthology.org/neurips/2023/dettmers2023neurips-qlora/

Reading focus: Focus on the effect of quantization and the distinction between LoRA and QLoRA.

VI. Format and Submission Requirements

- Team size: three students are recommended. Suggested preparation time: approximately two weeks after the topic is assigned.
- Submit a 6–10-page research-style report in Vietnamese on A4 paper (excluding references and appendices), organized as: Abstract, Introduction, Background/Paper Review, Method/Worked Example, Results and Discussion, Conclusion, and References.
- Prepare 8–12 content slides for a 15-minute seminar presentation followed by approximately 5 minutes of Q&A. Every team member must present and answer at least one question.
- Read all papers marked “Required” in this assignment. Explain the material in your own words and distinguish findings reported in the papers from examples or experiments developed by your team.

- Programming-based experiments are encouraged but not generally mandatory. A worked numerical example, an illustrative model, and an evidence-supported comparison are mandatory.
- Cite all figures, tables, datasets, equations, and AI tools used. Independently verify all claims suggested by AI-generated content.

VII. Seminar Assessment Rubric (100 points)

Assessment criterion	Requirements for full marks	Points
Problem statement and objectives	Establish relevant context, a clear research question, and an appropriate scope.	5
Reading and synthesis of research papers	Accurately explain the original papers' objectives, contributions, methods, and findings, with appropriate source comparison.	15
Explanation of algorithms and equations	Define all variables and assumptions, describe algorithmic steps, and provide at least one correct numerical example or diagram.	15
Evidence, comparison, or experimentation	Include an appropriate baseline or reference, verifiable data or examples, and clearly explained figures and tables.	10
Critical analysis and limitations	Discuss strengths, weaknesses, and conditions of applicability. Avoid overstating conclusions based on a small demonstration.	10
Slide quality and citations	Prepare 8–12 clear, legible slides, cite sources for figures/equations and relevant papers, and avoid verbatim copying.	10
Seminar presentation	Deliver a coherent presentation within 15 minutes, explain examples clearly, and ensure balanced participation.	15
Q&A and critical	Answer questions accurately, justify technical decisions,	20

discussion	and ensure that each member answers at least one question.	

Performance levels for each criterion: Excellent (100%), Good (75%), Satisfactory (50%), and Unsatisfactory (0–25%) of the available points, depending on the quality of evidence submitted.